Adabala Uday Raghavendra Kumar

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and ML practitioner with nearly 3 years of industry experience spanning data engineering, applied AI/ML, and quality assurance. Currently building robust data pipelines, RAG-powered document intelligence systems, and ML-based monitoring solutions for maritime IoT at Volteo Maritime. Hands-on expertise in LNG vessel telemetry, BOG modelling, CouchDB, Google Cloud Storage, and advanced retrieval-augmented generation (RAG) architectures. Published author on AI applications in maritime operations. Comfortable across the full data lifecycle — from ingestion and transformation to model deployment, evaluation, and quality validation.

EXPERIENCE

Data Scientist · Volteo Maritime · Full-time · *Jan 2025 – Present*

- Built and deployed an advanced RAG Document Assistant using LangChain, LlamaIndex, QdrantDB, NVIDIA NIM API, FastAPI, and Sentence Transformers — featuring hierarchical chunking, hybrid dense+sparse retrieval (HNSW + BM25), Reciprocal Rank Fusion, HyDE query rewriting, cross-encoder re-ranking, and NLI-based faithfulness scoring; demoed internally and iterated based on stakeholder feedback.
- Conducted structured RAG evaluation across 43 questions and three model versions (Gemma 3B, Latest Model, V1), designing a 5-layer evaluation framework with query drift tracking and confidence gating to identify regressions and guide model selection.
- Developed ML-based anomaly detection and data-quality models for real-time LNG vessel telemetry, applying statistical and time-series techniques to flag deviations and reduce manual monitoring effort by ~40%.
- Built and validated Boil-Off Gas (BOG) predictive models, applying data modelling and analysis techniques to improve cargo management accuracy and support operational decision-making for maritime clients.
- Designed and maintained real-time data pipelines to ingest and prepare sensor telemetry from LNG vessels as clean, model-ready inputs for downstream ML and analytics use cases.
- Prototyped a CouchDB-to-Power BI integration via FastAPI and Power BI Embedded, enabling real-time maritime analytics for non-technical stakeholders.

Software Engineer Trainee · Securden Technologies · Trainee · *Apr 2024 – Dec 2024*

- Tested early-stage features across the product suite and developed small internal features before moving on to build an internal AI chatbot.
- Validated features across three core enterprise security products: Privileged Access Management (PAM), Vault for MSPs, and Endpoint Privilege Manager (EPM).
- Executed comprehensive test strategies combining manual exploratory testing, REST API testing via Postman, and automation frameworks — achieving thorough coverage across functional, regression, and edge-case scenarios.
- Identified and escalated high-severity security misconfigurations in PAM workflows, preventing potential privilege-escalation vulnerabilities from reaching production.
- Authored detailed test plans, test cases, and defect reports; consistently met aggressive release timelines by coordinating closely with developers on bug triage.

Software QA Trainee · Securden Technologies · Trainee · *Aug 2022 – Sep 2022*

- Supported quality assurance activities as an early trainee, gaining foundational exposure to manual testing and defect reporting workflows.

PUBLICATIONS

AI-Powered Vessel Performance Monitoring in Maritime Operations
Medium (Volteo Maritime) · 2025
Adabala Uday Raghavendra Kumar
https://medium.com/@rag3kumar

Rose Plant Leaves: Disease Detection and Pesticide Management using CNN
IEEE Xplore · Jan 2022
Adabala Uday Raghavendra Kumar, Mohammed Khaleel, Paineni Gowtham, Raja P.
https://ieeexplore.ieee.org/document/9776898

EDUCATION

M.Tech in Artificial Intelligence & Data Science *Oct 2022 – Apr 2024*
SASTRA University
Coursework: Machine Learning · Deep Learning · Time Series Analysis · Reinforcement Learning · NLP · Statistics & Probability · R Language

B.Tech in Mechanical Engineering *Jul 2018 – Jul 2022*
SASTRA University
Coursework: Fluid Mechanics · Thermodynamics · Mechatronics · Manufacturing Technology · Mathematics & Statistics

TECHNICAL SKILLS

Languages: Python · R · SQL · C · C++

Data Engineering: ETL Pipeline Design · CouchDB · Google Cloud Storage (GCS) · Apache Spark · Data Quality Automation · Workflow Orchestration

AI / RAG: LangChain · LlamaIndex · QdrantDB · NVIDIA NIM API · Sentence Transformers · HyDE · Reciprocal Rank Fusion · Cross-Encoder Re-ranking · NLI Faithfulness Scoring

ML / AI: Supervised & Unsupervised Learning · Deep Learning · CNN · Transfer Learning · Reinforcement Learning · Time Series · NLP

Cloud & Infra: Google Cloud Platform (GCP) · Vertex AI · Google Cloud Storage · FastAPI · Ollama · Docker (basic)

LLM Providers: NVIDIA NIM · Google Gemini · Ollama (Gemma, LLaMA) · Anthropic Claude API

Tools & Platforms: Power BI · Power BI Embedded · Streamlit · Jupyter Notebook · Google Colab · Postman

QA & Testing: Manual Testing · API Testing (Postman) · Test Planning · Regression Testing · Defect Lifecycle Management

Languages (Human): English · Telugu · Hindi · Tamil (spoken)